

LLMs and their Multilingualism

François Yvon

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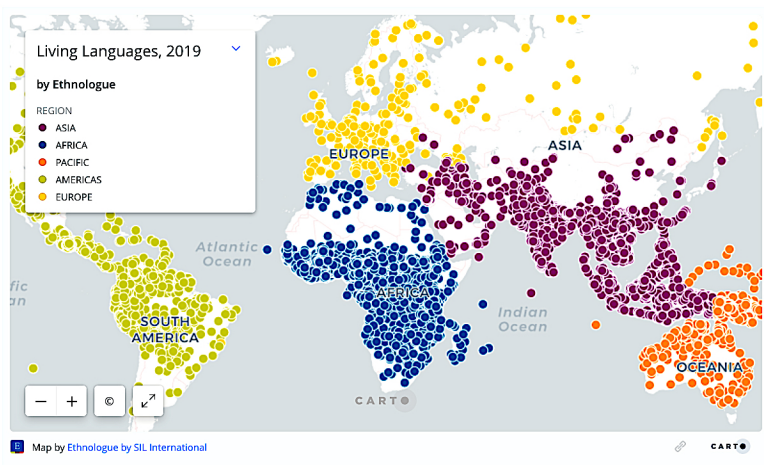


2025 / 2026

Part I

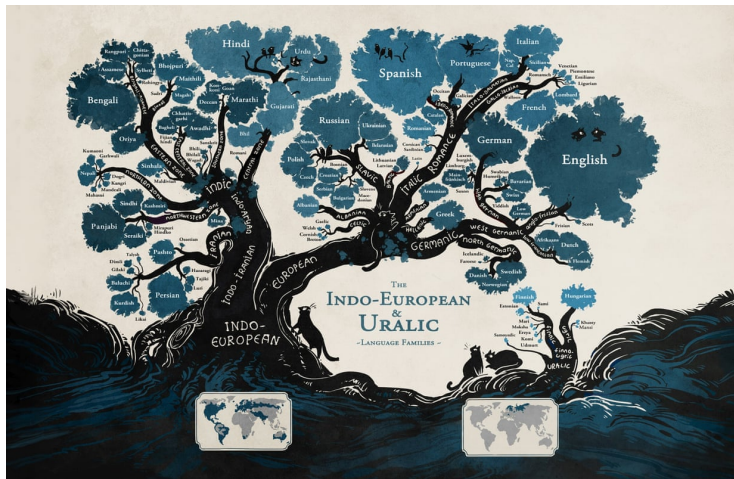
Living in a Multilingual World

Living in a Multilingual World



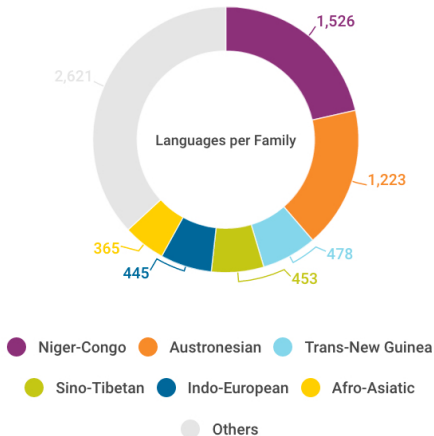
<https://www.ethnologue.com/guides/how-many-languages>

Living in a Multilingual World



 A Language Family Tree

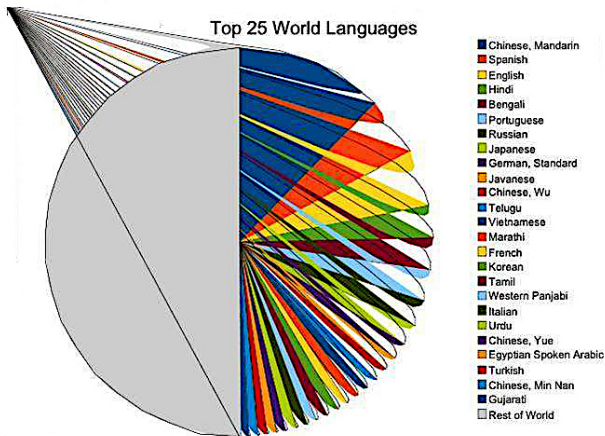
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<https://www.ethnologue.com/guides/largestfamilies>

Living in a Multilingual World

Some languages are more equal than others



 <https://terralingua.org/>

Living in a Multilingual World

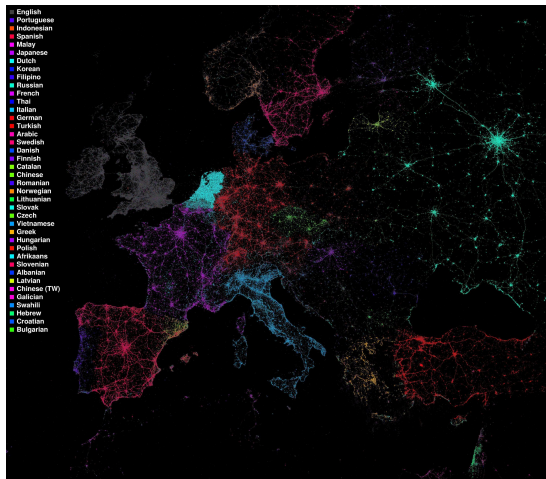
Measuring Language Use on the Internet

LANGUES ¹⁰	POP. MOND.	TRAF.	INTER.	USAG.	CONT.	INTERF.	INDEX	POP. CONN.	PUISS.	CAPAC.	GRAD.
Anglais	13,0 %	37,9 %	15,2 %	27,9 %	38,7 %	21,6 %	17,6 %	64,2 %	26,5 %	2,04	1,74
Chinois	14,7 %	7,9 %	17,6 %	5,5 %	8,2 %	25,0 %	19,2 %	65,6 %	13,9 %	0,94	0,79
Espagnol	5,2 %	10,8 %	6,9 %	11,7 %	5,4 %	9,8 %	7,5 %	72,5 %	8,7 %	1,66	1,26
Français	2,6 %	2,7 %	3,0 %	3,8 %	5,4 %	4,3 %	3,2 %	64,9 %	3,7 %	1,45	1,23
Hindi	5,8 %	4,9 %	4,3 %	3,2 %	0,3 %	4,1 %	3,8 %	41,2 %	3,4 %	0,59	0,79
Portugais	2,5 %	1,5 %	3,1 %	5,5 %	3,3 %	3,9 %	3,0 %	68,3 %	3,4 %	1,36	1,09
Russe	2,5 %	1,8 %	3,6 %	2,3 %	3,4 %	3,9 %	3,8 %	78,3 %	3,1 %	1,26	0,88
Arabe	3,5 %	2,3 %	3,8 %	3,0 %	2,1 %	4,2 %	2,9 %	59,0 %	3,0 %	0,86	0,80
Allemand	1,3 %	1,3 %	2,1 %	2,0 %	5,9 %	2,8 %	3,0 %	87,7 %	2,8 %	2,17	1,36
Japonais	1,2 %	2,0 %	2,1 %	1,8 %	3,5 %	2,8 %	3,0 %	92,6 %	2,5 %	2,06	1,22
Malais	2,4 %	0,9 %	2,2 %	2,8 %	0,8 %	1,9 %	2,0 %	51,0 %	1,8 %	0,74	0,80
Italien	0,7 %	0,5 %	0,9 %	1,0 %	3,4 %	1,2 %	1,2 %	75,7 %	1,4 %	2,09	1,51
Turc	0,9 %	1,0 %	1,2 %	1,6 %	0,9 %	1,4 %	1,2 %	78,0 %	1,2 %	1,46	1,02
Coréen	0,8 %	0,9 %	0,9 %	1,0 %	0,8 %	1,1 %	0,9 %	64,7 %	1,0 %	1,22	1,03
Bengali	2,6 %	1,2 %	1,2 %	1,1 %	0,3 %	0,7 %	0,8 %	24,5 %	0,9 %	0,35	0,77
RESTE	40,4 %	22,3 %	31,9 %	25,9 %	17,6 %	11,3 %	26,9 %		22,7 %		
TOTAL	100 %	100 %	100 %	100,0 %	100 %	100 %	100 %	54,9 %	100 %		

Obs. Francophonie, Rapport 22 <https://observatoire.francophonie.org/>

Living in a Multilingual World

Measuring Language Use is Difficult

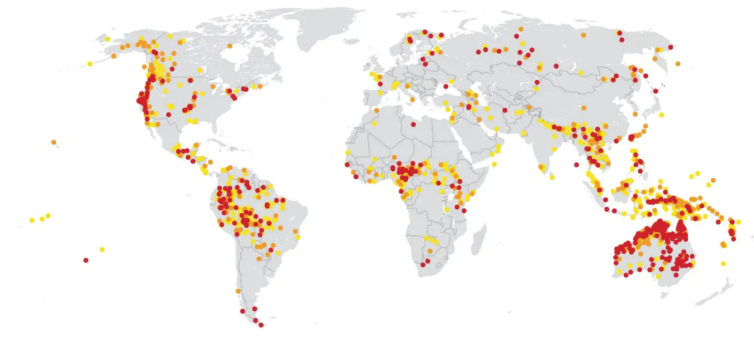


📄 (C) E. Fisher - CC-BY

Living in a Multilingual World

Language diversity, a treasure to cherish

● Critically endangered ● Seriously endangered ● Endangered



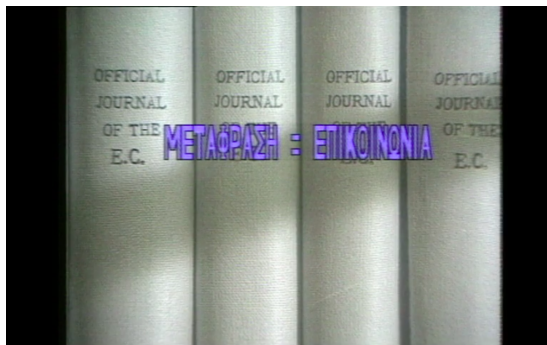
Dynamic maps <https://www.ethnologue.com/>

Living in a Multilingual World



👉 160+ writing systems <https://www.unicode.org/iso15924/iso15924-codes.html>

Multilingualism: a Treasure that also Costs



📺 <https://audiovisual.ec.europa.eu/en/video/I-000267>

Multilingualism: a Treasure that also Costs

A very large translation agency: European institutions



Caution: figures from 2011. Excluding Translator-Lawyers and Interpreters.

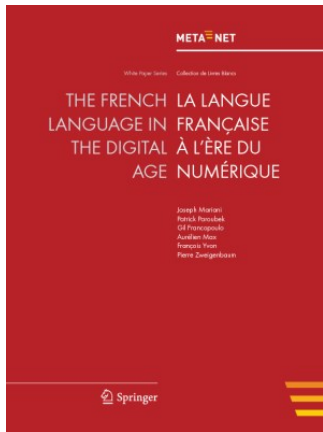
Also: corpora (JRC Acquis, Europarl), translation memories (Euramis), Terminology (IATE)

Alt. view: the true cost of multilingualism = 1 coffee / European / year

📄 https://ec.europa.eu/info/departments/translation_en

Multilingualism: a Treasure that also Costs

How can language technologies help?



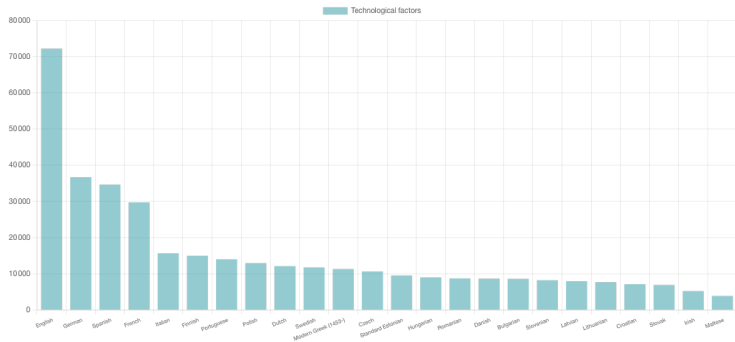
2010



2022

Multilingualism: a Treasure that also Costs

Can language technologies help ?



 <https://live.european-language-grid.eu/catalogue/dashboard>

Reasons why NLPers should care

Inspiration:

📖 <https://runder.io/nlp-beyond-english/>

- **Political / societal:** breaking the digital divide / equal access to education, justice, culture, knowledge; mitigating biases such as disfunctional technologies (accents, sociolects, people with disabilities)
- **Economic:** doing business in a multilingual world requires ... multilingualism (sales sites, contracts, customer care, procurement, certificates, instruction manuals, etc)
- **Cultural / Historical:** languages are disappearing at an alarming pace; language technologies can help collect and document world's diversity and heritage
- **Linguistic:** English and the like display a small fraction of linguistic phenomena (fixed word order, a-tonal phonology, restricted morphology, etc); studying more languages increases knowledge of linguistic phenomena and their distribution
- **Cognitive:** human beings learn language, no matter how diverse, with the same organs
- **Machine Learning:** generalization is also cross-lingual generalization - new issues: small samples, training distributions, etc.
- **Computational:** one model to train, deploy, maintain, update.

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Political Motivations for Multilingual Technologies

The right to education, justice, culture, trade, in one's own language

What the treaties say, and how to enforce it

Article 21 – Non-discrimination

1. Any discrimination based on any ground such as sex, race, colour, ethnic or social origin, genetic features, *language*, religion or belief, political or any other opinion, membership of a national minority, property, birth, disability, age or sexual orientation shall be prohibited.

Article 22 – Cultural, religious and linguistic diversity

The Union shall respect cultural, religious and *linguistic diversity*.

Article 41 – Right to good administration

4. Every person *may write* to the institutions of the Union in one of the languages of the Treaties and *must have an answer* in the same language

Political Motivations for Multilingual Technologies

The right to education, justice, culture, trade, in one's own language

*The European Parliament (...) notes the **deepening digital divide between widely used and lesser-used languages**, and the increasing digitalisation of European society, which is **leading to disparities in access to information, particularly for the low-skilled, the elderly, people on low incomes and people from disadvantaged backgrounds**; stresses that **making content available in different languages would reduce inequality**;*

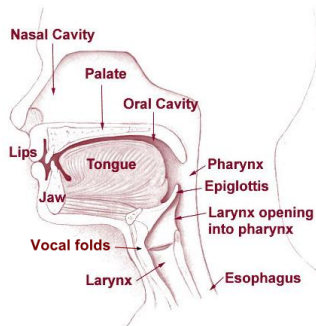
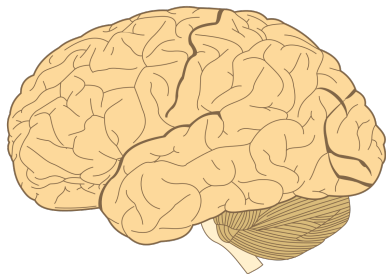
(...) calls on the Commission and the Member States to develop strategies and policy action to facilitate multilingualism in the digital market;

(...) emphasises that the development of language technology will facilitate the subtitling, dubbing and translation of video games and software applications into minority and lesser-used languages;

 https://www.europarl.europa.eu/doceo/document/A-8-2018-0228_EN.pdf

Linguistic Motivations for Multilingualism in NLP

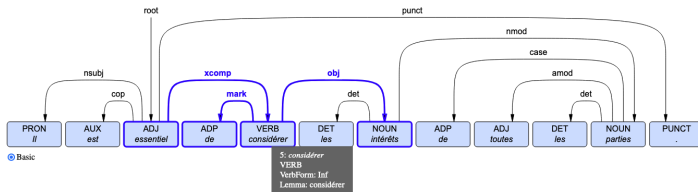
Language organs are universal - can computational models help their analysis?



Linguistic Motivations for Multilingualism in NLP

sent_id = Europar.550_00063

text = Il est essentiel de considérer les intérêts de toutes les parties.

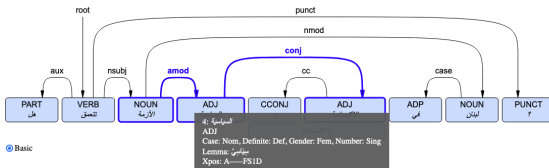


newpar id = assabah.20041022.0016.p2

sent_id = assabah.20041022.0016.p2u1

text = هل تتعلق الأزمة السياسية والاقتصادية في لبنان؟

orig_file_sentence ASB_ARB_20041022.0016#2



Yet, languages are amazingly diverse

Machine Learning Motivations for Multilingual NLP

Comparative Hardness of Language Modeling with n-grams

Language models: a basic NLP technology

$$\begin{aligned} P_{\theta}(w_1 \dots w_L) &= \prod_{t=1}^T P(w_t | w_1 \dots w_{t-1}; \theta) \\ &= \prod_{t=1}^T P_{\text{ngram}}(w_t | w_{t-n+1} \dots w_{t-1}; \theta) \\ &= \prod_{t=1}^L P_{\text{NN}}(w_t | w_{<t}; \theta) \end{aligned}$$

$\forall$ context h , $P(w|h; \theta)$ is a discrete distribution over the set of possible units

Compare candidate sentences (Automatic Speech Recognition or Machine Translation), generate new random texts, etc.

Machine Learning Motivations for Multilingual NLP

Comparative Hardness of Language Modeling with n-grams

Evaluating language models: perplexity

$$\text{PPL}(M_{\theta}) = 2^{\frac{1}{T} \log_2 P(w_{[1:T]} | \theta)} = P(w_{[1:T]} | \theta)^{-\frac{1}{T}} \text{ [nats]}$$

- PLL is related to the cross-entropy between actual source and model M_{θ}
- $\text{PPL}(\text{Unif}) = 2^{\frac{1}{T} \log_2 P(w_{[1:T]} | M)} = 2^{\frac{1}{T} T \log_2 (1/n_W)} = n_W$

Machine Learning Motivations for Multilingual NLP

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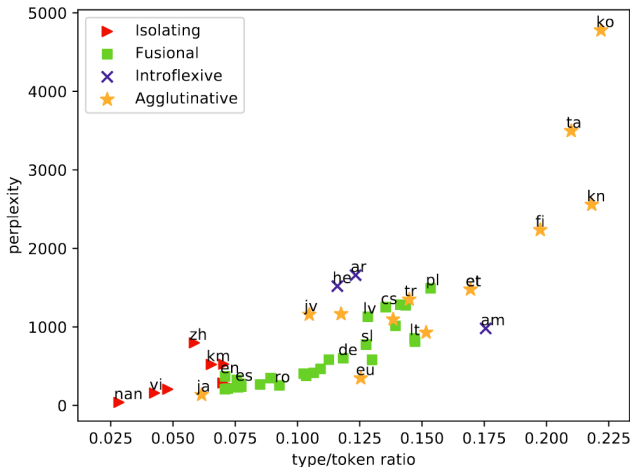
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Is M a better model for L_1 than for L_2 ?

- comparable training / testing texts $w_1 \dots w_T$ (same domains, topics, genres)
- a comparable unit / event set $\mathcal{V}$ for $P()$
- comparable normalizer (T)
- comparable strategies for Out-of-Vocabulary (OoV) words

Machine Learning Motivations for Multilingual NLP

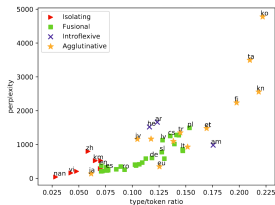
Comparative Hardness of Language Modeling with n-grams



📄 (Gerz et al, 2018) <https://www.aclweb.org/anthology/D18-1029.pdf>

Machine Learning Motivations for Multilingual NLP

Comparative Hardness of Language Modeling with n-grams



*Importantly, our large-scale multilingual LM study strongly indicates that due to diverse typological profiles, **certain languages and language groups are inherently more complex to language-model** when relying on established statistical models, even when such models are constructed as widely applicable and (arguably) language-agnostic.*

📖 (Gerz et al, 2018) <https://www.aclweb.org/anthology/D18-1029.pdf>

Methods and Resources in Multilingual NLP

- Machine Translation [1953-]
 - Cross-Lingual Lexical Representations [2013-]
 - **Multilingual language models** (mBERT, mBART, mT5, XLM-R, etc) [2016-]
-
- List of languages and language families  <https://glottolog.org>
 - Parallel and comparable corpora  <https://opus.nlpl.eu>
 <https://wikipedia.org>
 - Bilingual dictionaries  <https://panlex.org>
 - Comparative / **typological language documentation**  <https://wals.org>

Methods and Resources in Multilingual NLP

Parallel Paragraphs

In the gayest and happiest spirits she set forward with her father; not always listening , but always agreeing to what he said. They arrived. It is Frank and Miss Fairfax , said Mrs. Weston. I was just going to tell you of our agreeable surprize in seeing him arrive this morning. He stays till to-morrow , and Miss Fairfax has been persuaded to spend the day with us.

Elle partit avec son père, le visage souriant; elle n'écoutait pas toujours, mais elle acquiesçait de confiance. Ils arrivèrent. – C' est Frank et Mlle Fairfax, dit aussitôt Mme Weston. – J'allai justement vous faire part de l'agréable surprise que nous avons eue en le voyant arriver. Il reste jusqu'à demain et Mlle Fairfax a bien voulu, sur notre demande, venir passer la journée .

📖 Jane Austen, Emma <https://www.janeausten.org/emma/chapter-54.asp>

Methods and Resources in Multilingual NLP

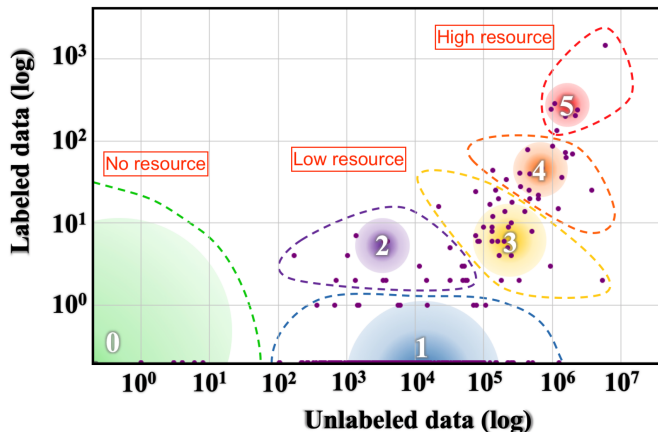
Parallel Sentences

In the gayest and happiest spirits she set forward with her father;	Elle partit avec son père, le visage souriant;
not always listening, but always agreeing to what he said;	elle n' écoutait pas toujours, mais elle acquiesçait de confiance.
They arrived .	Ils arrivèrent .
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He stays till tomorrow, and Miss Fairfax has been persuaded to spend the day with us .	Il reste jusqu'à demain et Mlle Fairfax a bien voulu, sur notre demande , venir passer la journée.

Aligning Sentence in Parallel Documents is Easy

Challenges in Multilingual NLP

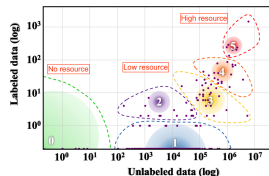
Resource Unbalance



📖 Joshi et al. [2020] <https://arxiv.org/pdf/2004.09095.pdf>

Challenges in Multilingual NLP

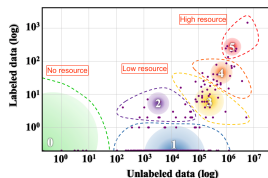
Resource Unbalance



Cls	5 example languages	#Lang	#Spkr	% Lang
0	Dahalo, Warlpiri, Popoloca, Wallisian, Bora	2191	1.2B	88.38%
1	Cherokee, Fijian, Greenlandic, Bhojpuri, Navajo	222	30M	5.49%
2	Zulu, Konkani, Lao, Maltese, Irish	19	5.7M	0.36%
3	Indonesian, Ukranian, Cebuano, Afrikaans, Hebrew	28	1.8B	4.42%
4	Russian, Hungarian, Vietnamese, Dutch, Korean	18	2.2B	1.07 %
5	English, Spanish, German, Japanese, French	7	2.5B	0.28%

📖 Joshi et al. [2020] <https://arxiv.org/pdf/2004.09095.pdf>

Challenges in Multilingual NLP



“Low-Resource” vs “High-Resource” languages are Slippery Concepts

- some LR languages have many speakers and many resources
- equal technological support for all languages ? Ask speaker communities!
- LTs imply normalisation, not always desirable
- against linguistic extractivism

Challenges in Multilingual NLP

Strengths and Shortcomings of the MT-based strategy

Facebook

Introducing the First AI Model That Translates 100 Languages Without Relying on English

- Improving (English-centric) MT benefits all technologies
- NLLB only covers $\sim 10^4$ translation pairs, out of 50 million
- MT far from solved even for close languages
- requires resources that do not (and will never) exist

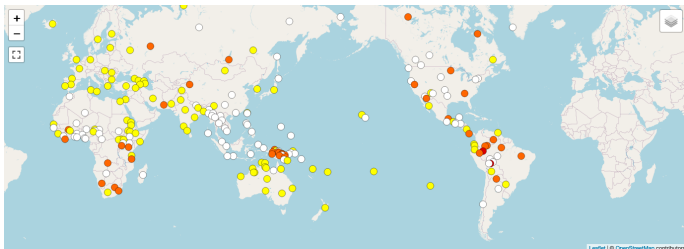
📄 Costa-jussà et al. [2024]

<https://about.fb.com/news/2020/10/first-multilingual-machine-translation-model/>

Challenges in Multilingual NLP

Language difference and similarities

- designing language-agnostic models
- exploiting language relatedness and linguistic priors
- in spite of a tremendous diversity at all levels (phonology, morphology, syntax, semantics)



Languages with no past tense, regular past tense, 2-3 remoteness, 4+ remoteness.

Feature 66A <https://wals.info>

Challenges in Multilingual NLP

New (interesting) problems: language contact and code-switching

Code-Switching (CS) denotes the alternation of two languages within a single utterance. It is a common communicative phenomenon that occurs in multilingual communities during informal spoken and written interactions.

This morning mi hermano y yo fuimos a comprar some milk
Son post, j'l'ai liké comme un ouf

New tasks:

- CS language identification
- CS language transcription and analysis
- CS language translation
- CS generation (hard)

☞ Sitaram et al. [2019] <https://arxiv.org/pdf/1904.00784.pdf>

The Best Reason to Pay Attention

All comtemporary models are multilingual

Hugging Face Search models, datasets, users...

Models 6,557 Filter by name

Full-text search Inference Available 11 Sort: Trending

Filter Languages by name Reset Languages

English Chinese French Spanish German Japanese Korean Portuguese Italian Russian Hindi Arabic Thai multilingual Turkish Indonesian Vietnamese Polish Dutch Romanian Swedish Ukrainian Persian Czech Finnish Bengali Nepali Hebrew Greek Danish Malay Tamil Hungarian Urdu Bulgarian Catalan French Telugu Marathi Serbian Norwegian Swahili Gujarati Slovak Estonian Slovenian Burmese Malayalam Croatian Tagalog Lithuanian Galician Khmer Latvian Kannada Basque Icelandic Panjabi Lao Amharic Mongolian Afrikaans Kazakh Georgian Assamese Hausa Armenian Macedonian Welsh Belarusian Sinhala English Azerbaijani Javanese Uzbek Sundanese Irish Yoruba Albanian Latin Bosnian Somali Maltese Thai Sanskrit Spanish code Russian Oriya Arabic

+ 4772 languages

**January 26:
6500+ models
in 4772 languages**

Models 6,557

- deepseek-ai/DeepSeek-OCR
Image-Text-Text · 3B · Updated Nov 4, 2025 · 3.17M · 3.07k
- VAGOSolutions/SauerkrautLM-Translator-LFM2.5-1.2B
Translation · 1B · Updated 3 days ago · 158 · 11
- rednote-hilab/dots.ocr
Image-Text-Text · 3B · Updated Oct 31, 2025 · 445k · 1.18k
- Octen/Octen-Embedding-8B
Sentence Similarity · 8B · Updated about 19 hours ago · 4.85k · 12
- nvidia/llama-embed-nemotron-8b
Feature Extraction · 8B · Updated Jan 26, 2026 · 3.49M · 487
- sentence-transformers
Sentence Similarity · 1B · Updated Jan 26, 2026 · 2.95M · 1.12k
- microsoft/Phi-4
Text Generation · 4B · Updated Jan 26, 2026 · 77.7k · 451
- openbmb/MiniCPM
Image-Text-Text · 7B · Updated Jan 26, 2026 · 59 · 6
- intfloat/multilingual-e5-small
Sentence Similarity · 0.1B · Updated Feb 17, 2025 · 1.67M · 265
- pipecat-ai/smart-turn-v3
Voice Activity Detection · Updated 9 days ago · 112
- nvidia/llama-nemotron-embed-v1-1b-v2
Feature Extraction · 2B · Updated 23 days ago · 1.98k · 14
- PaddlePaddle/PaddleOCR-VL
Image-Text-Text · 1.0B · Updated Dec 11, 2025 · 12.5k · 1.49k
- LEMAS-Project/LEMAS-TTS
Text-to-Speech · Updated 6 days ago · 30 · 10
- TheStageAI/thewhisper-large-v3-turbo
Automatic Speech Recognition · 0.8B · Updated 5 days ago · 1.78k · 14
- BAAI/bge-reranker-v2-m3
Text Classification · 0.6B · Updated Jun 24, 2024 · 2.8M · 861
- FacebookAI/xlm-roberta-large
Text Classification · 1.3B · Updated Jan 26, 2026 · 3.49M · 487
- intfloat/multilingual-e5-large-instruct
Feature Extraction · 0.6B · Updated Jul 10, 2025 · 1.31M · 600
- FabioSarracino/VibeVoice-Large-08
Text-to-Audio · 8B · Updated Oct 1, 2025 · 2.75k · 76
- microsoft/Phi-3.5-mini-instruct
Text Generation · 4B · Updated Dec 10, 2025 · 329k · 946

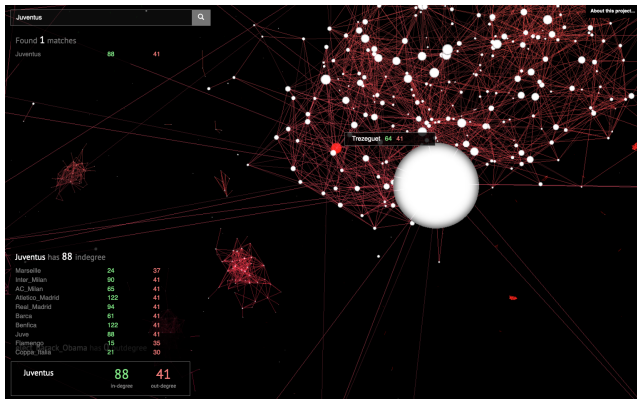
Part II

Training mLLMs

Pretrained Language Models are a Wonder

Modern Lexical Representations : Word Embeddings

Map words w_t in context c onto vector-space representations x as $x = f_{\theta}(w_t; c)$.
 $f_{\theta}()$ is a neural net e.g., Transformer [Vaswani et al., 2017] with parameters θ .



The structure of latent spaces: related words / sentences are close

Pretrained Language Models are a Wonder

Learning θ

Gigantic collections of texts, simple **auxiliary tasks** with **natural annotations**

- 1 Predict next word given prefix : **pure decoder**

Longtemps je me suis couché [mask] ... $\Rightarrow$ train $P_{\theta}(w_t|w_{<t})$

GPT-2 [Radford et al., 2019], OPT [Zhang et al., 2022], GPT-J [Wang and Komatsuzaki, 2021], etc

- 2 Predict missing word given bidirectional context: **pure encoder**

Longtemps je me suis couché de [mask] bonne heure. $\Rightarrow$ train $P_{\theta}(w_t|w_{-t})$

BERT [Devlin et al., 2019], Roberta [Liu et al., 2019], CamemBERT [Martin et al., 2020], etc.

- 3 Denoising sequence-to-sequence: **encoder-decoder**

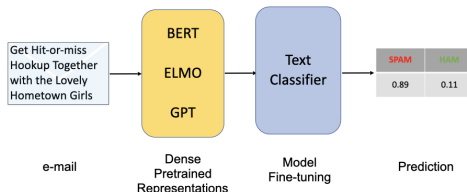
Longtemts je couché suis de bnone || Longtemps je me suis couché de bonne heure.
 $\Rightarrow$ train $P_{\theta}(w|\tilde{w})$

BART [Lewis et al., 2020], T5 [Raffel et al., 2020], etc.

Pretrained Language Models are a Wonder

Using θ

- 1 Learn + fine-tune **task-adapted** model $h_{\phi, \theta}(w; c) = h_{\phi}(f_{\theta}(w; c))$



- 2 With causal LMs, multi-purpose text generation via **prompting**

Pretrained Language Models are a Wonder

Using θ

- 1 Learn + fine-tune **task-adapted** model $h_{\phi, \theta}(w; c) = h_{\phi}(f_{\theta}(w; c))$
- 2 With causal LMs, multi-purpose text generation via prompting

GEN Of course. In Chorukor, Monday is ilopagar, Tuesday ilopager, ...
Wednesday ilopagur, Thursday ilopagir ...

Q&A Answer this: What are the birth date and place of Ludvík Vaculík? ...
23 July 1926, in Brumov, Moravia

SA "This Czech writer has written some the most wonderful French novels."
is a positive comment ? ... [Yes | No]

Pretrained Language Models are a Wonder

Using θ

- 1 Learn + fine-tune **task-adapted** model $h_{\phi,\theta}(w;c) = h_{\phi}(f_{\theta}(w;c))$
- 2 With causal LMs, multi-purpose text generation via prompting

- ☺ LM pre-training is mostly unsupervised
- ☺ Instruction-following can be learned or fine-tuned
- ☺ Instruction can include exemplars for few-shot in-context learning
- ☺ Effective recipes for post-training “alignment” and “safeguarding”
- ☺ Emerging “reasoning” abilities
- ☺ Extremely successful for English, French, German, etc.
- ☺ Annotations, corpus and model size matter

Can we generalize to all Languages?

Pretrained Language Models are a Wonder

Using θ

- 1 Learn + fine-tune **task-adapted** model $h_{\phi,\theta}(w;c) = h_{\phi}(f_{\theta}(w;c))$
- 2 With causal LMs, multi-purpose text generation via prompting

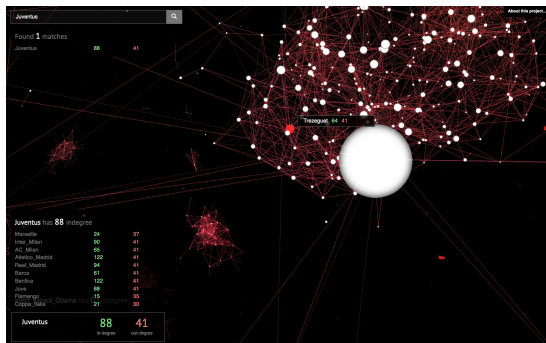
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Can we generalize to all Languages?

Multilingual Pretrained Language Models - mPLMs

Aligning Multilingual Word Embeddings [Ruder et al., 2019]

Map words w onto vector-space representations x in context c as $x = f_{\theta}(w; c)$
 $f_{\theta}()$ a complex neural net with parameters θ .



Neighbours are semantically similar **cross-linguistically**

Multilingual Pretrained Language Models - mPLMs

Multilingual Tokenization

[Using XLM-Roberta]

_Tous _les _être s _humain s _na issent _libre s _et _ég aux _en _digni té _et _en
_droits . _Ils _sont _do u és _de _raison _et _de _conscience _et _doivent _agir _les
_uns _en vers _les _autres _dans _un _esprit _de _frater n ité .

_Všichni _lidé _rod í _se _svobod ní _a _sobě _rov ní _co _do _d ů stoj nosti _a _práv
_Jsou _na dán i _rozum em _a _s vědomí m _a _mají _spolu _jedna t _v _du chu
_brat r ství .

_Tutti _gli _esse ri _umani _na scono _liberi _ed _e gu ali _in _digni tà _e _diritti
_Es si _sono _do tati _di _ragione _e _di _coscienza _e _devono _agir e _gli _uni
_verso _gli _altri _in _spirito _di _fra tella nza .

_Ο _λοι _οι _άνθρωποι _γενν ι ούνται _ελεύθερο ι _και _ί σο ι _στην _αξι οπ ρέ πει
_α _και _τα _δικαιώματα . _Εί ναι _προ ικ ισμέ νοι _με _λογική _και _συνείδηση , _και
_οφείλ ουν _να _συ μπερι φέρ ονται _μεταξύ _τους _με _πνεύμα _αδελφ οσύνη ς .

Low-level, language independent segmentations
From sequence of bytes, codepoints, subwords

Multilingual Pretrained Language Models - mPLMs

Multilingual Tokenization

[Using XLM-Roberta]

_Tous _les _être s _humain s _na issent _libre s _et _ég aux _en _digni té _et _en
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_Es si _sono _do tati _di _ragione _e _di _coscienza _e _devono _agir e _gli _uni
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Low-level, language independent segmentations
From sequence of bytes, codepoints, subwords

Multilingual Pretrained Language Models - mPLMs

PLMs model **sequences of tokens** e.g.,

$$P([u_1 \dots u_T] | \theta)$$

Whether they are “monolingual” or “multilingual” is irrelevant

End of Class - Any Question ?

Multilingual Pretrained Language Models - mPLMs

Learning multilingual θ

Multilingual Mixtures of corpora, simple **auxiliary tasks** with **natural annotations**:

- 1 Predict next word given prefix : **pure decoder**.
XGLM [Lin et al., 2022], BLOOM [Scao et al., 2022] mGPT [Shliazhko et al., 2024], etc
- 2 Predict missing word given full context: **pure encoder**.
mBERT [Devlin et al., 2019], XLM-R [Conneau et al., 2020]
- 3 Denoising sequence to sequence: **encoder-decoder**
mBART50 [Liu et al., 2020b], mT5 [Xue et al., 2021], etc

+ **Auxiliary objectives** to bridge between languages:
parallel corpora, bilingual dictionaries, synthetic code-switched data, transliteration
[Conneau et al., 2020, Hu et al., 2021, Chi et al., 2021], etc.

Multilingual Pretrained Language Models - mPLMs

Using multilingual θ

- 1 mPLMs are polyglot: **monolingual multitask learners**, in all languages
- 2 Learn / fine-tune task-adapted model $h_{\phi, \theta}(w; c) = h_{\phi}(f_{\theta}(w; c))$ on L_1 , **perform zero-shot X-lingual transfer to L_2**
- 3 Multilingual text generation with prompting

Translate into French

“ By the end of the year, we will have seven new pharmacists. ”:

D'ici la fin de l'année, nous aurons sept nouveaux pharmaciens.

Multilingual Pretrained Language Models - mPLMs

Using multilingual θ

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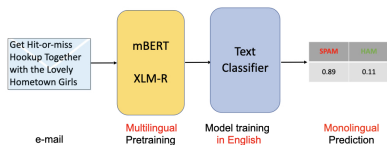
- ☺ Hardly more difficult than monolingual
- ☺ Bring less resourced languages up to steam

Part III

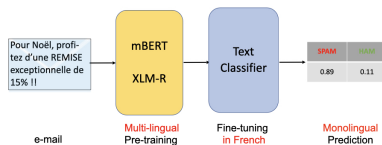
Dissecting New Abilities of mPLMs

The New Abilities of Multilingual PLMs

One model handling multiple languages



English spam filter

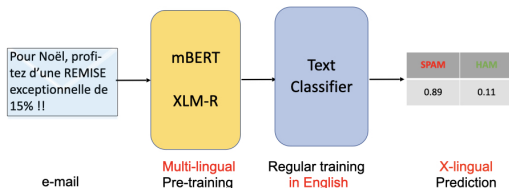


French spam filter

Effective for many languages & tasks [Pires et al., 2019, Wu and Dredze, 2020]

The New Abilities of Multilingual PLMs

Knowledge transfer between languages



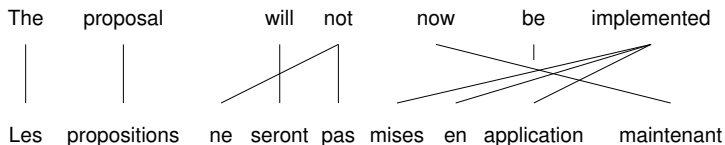
Multilingual zero-shot spam filter

- ☺ No foreign training data required
- ☺ Effective for many languages and tasks: X-GLUE [Liang et al., 2020], X-TREME* [Hu et al., 2020, Ruder et al., 2021, 2023]

The New Abilities of Multilingual PLMs

Multilingual Embeddings Align Lexical Representations

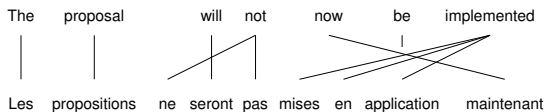
Aligning words, an **old, ill-posed and vexing** problem



The New Abilities of Multilingual PLMs

Multilingual Embeddings Align Lexical Representations

Aligning words, an **old, ill-posed and vexing** problem



A simple alignment algorithm:

- Each word is a node in a bipartite graph
- X-lingual links are valued with similarity
- Maximal weight matching a.k.a Hungarian Wedding $\Rightarrow$ **partial 1-1 alignment**
- Works for all language pairs !

Method	ENG-CES		ENG-DEU		ENG-FAS	
	F_1	AER	F_1	AER	F_1	AER
IBM2	.76	.25	.71	.29	.57	.43
IBM4	.75	.26	.77	.23	.51	.49
eflomal	.85	.15	.77	.23	.61	.39
fastText	.70	.30	.60	.40	.50	.50
mBERT[8]	.87	.13	.79	.21	.67	.33
XML-R[8]	.87	.13	.79	.21	.70	.30

Word alignment, Argmax method

[Jalili Sabet et al., 2020]

The New Abilities of Multilingual PLMs

Multilingual Text Generation: Translation

If the French says: Ça n'arrête pas, nous sommes bien placés pour le savoir..
 then the English should say: It doesn't stop, we know that well...

Model	En-Fr	Fr-En
GPT-2 [1,5b] [Radford et al., 2019]	5	11.5
GPT-3 [175b] [Brown et al., 2020]	21.2	25.2
PALM [540B] [Chowdhery et al., 2024]	38.5	41.1
BLOOM [176B] [Scao et al., 2022]	??	??
2022 SOTA [Liu et al., 2020a]	44.1	-

Evaluation use **BLEU** - a surface-based metric

Other tasks : X-lingual QA and Summarization, IR, etc

The New Abilities of Multilingual PLMs

Understanding MT performance of BLOOM

Methods

- Multiple prompts, in multiple languages
- Multiple datasets
- Dozens of language pairs (e.g. from Flores)
- Multiple metrics

Main takeaways

- **failure of 0-shot learning** from “language hallucinations” to “off-target” generation
- **effectiveness of k -shot learning**
- “Free-rider effects” and language similarities
- low MT quality for LR languages

Src ↓	Trg →	ar	en	es	fr	zh
ar	BLOOM	–	40.3	23.3	33.1	17.7
	M2M	–	25.5	16.7	25.7	13.1
en	BLOOM	28.2	–	29.4	45.0	26.7
	M2M	17.9	–	25.6	42.0	19.3
es	BLOOM	18.8	32.7	–	24.8	20.9
	M2M	12.1	25.1	–	29.3	14.9
fr	BLOOM	23.4	45.6	27.5	–	23.2
	M2M	15.4	37.2	25.6	–	17.6
zh	BLOOM	15.0	30.5	20.5	26.0	–
	M2M	11.6	20.9	16.9	24.3	–

(a) High-resource language pairs.

Src ↓	Trg →	en	fr	hi	id	vi
en	BLOOM	–	45.0	27.2	39.0	28.5
	M2M	–	42.0	28.1	37.3	35.1
fr	BLOOM	45.6	–	18.5	31.4	32.8
	M2M	37.2	–	22.9	29.1	30.3
hi	BLOOM	35.1	27.6	–	–	–
	M2M	27.9	25.9	–	–	–
id	BLOOM	43.2	30.4	–	–	–
	M2M	33.7	30.8	–	–	–
vi	BLOOM	38.7	26.8	–	–	–
	M2M	29.5	25.8	–	–	–

(b) High→mid-resource language pairs.

Packing Multiple Languages into a single PLM

Linguistic observations

- Lexical overlap not necessary

L1: Longtemps je me suis couché de bonne heure .

L2: Mpohufnqt kf nf tvjt dpvdiê ef cpooof ifvsf !

But it certainly helps

- Vocabulary alignment helps transfer
- Language typological similarity helps transfer
- **Transfer breaks** with reversal or words shuffle

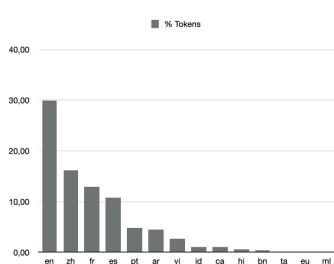
L1: Longtemps je me suis couché de bonne heure .

L2: . heure bonne de couché suis me je longtemps

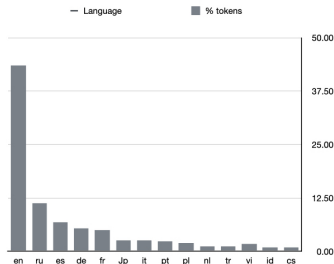
[Pires et al., 2019, Dufter and Schütze, 2020, Deshpande et al., 2022, Philipppy et al., 2023], etc

Statistical observations: mPLMs model data mixtures

Language Training Distribution can be **very unbalanced**



Languages in BLOOM



Languages in mT5

- transfer happens in very unbalanced models [Schäfer et al., 2024]
- better balancing through “squashing” - upsampling LR languages $\sim P(l)^\alpha$, $\alpha < 1$
- better balancing through **UniMax** sampling [Chung et al., 2023]:
Given compute budget: **max corpus repeats** for LR, **uniform sampling** for HR

Packing Multiple Languages into a single PLM

- ⊗ Boosting and breaking X-lingual transfer
- ⊗ Impact of language distributions
- Impact of model size & architecture
- Impact of number of languages - the “**curse of multilingualism**” [Conneau et al., 2020]
- Same questions for post-training

All this requires **multilingual evaluations & comparison**

Multilingualism: a Blessing or a Curse ?

A Direct Measure of Transfer Effects [Longpre et al., 2025]

$$\text{BTS}_{s \rightarrow t} = \frac{2d_{\text{mono}} - \sigma_{\text{bi}}(\ell_t(d_{\text{mono}}))}{d_{\text{mono}}}$$

where:

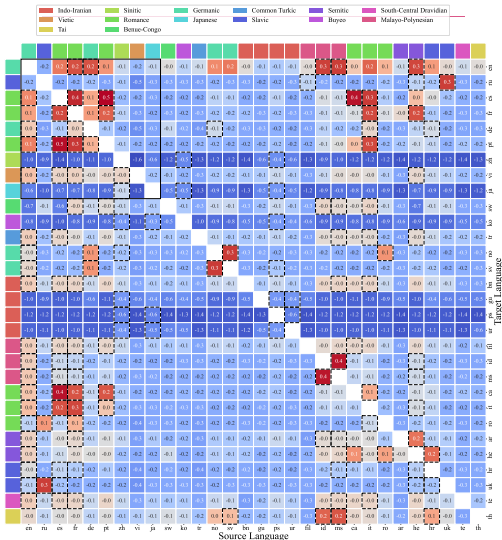
- d_{mono} : target monolingual token budget
- $\ell_t(d_{\text{mono}})$: loss achieved when optimally training a monolingual target 2B model with d_{mono} tokens
- $\sigma_{\text{bi}}(\ell)$: number of bilingual tokens equally chosen in (s, t) to reach loss ℓ

Interpretation

- $\text{BTS}_{s \rightarrow t} > 0$: achieving $\ell_t(d_{\text{mono}})$ requires less than d_{mono} target tokens: **positive transfer**
- $\text{BTS}_{s \rightarrow t} > 0$: achieving $\ell_t(d_{\text{mono}})$ requires more than d_{mono} target tokens: **negative transfer a.k.a interference**

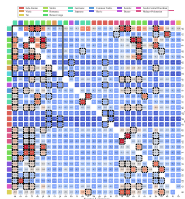
Multilingualism: a Blessing or a Curse ?

A Direct Measure of Transfer Effects [Longpre et al., 2025]



Multilingualism: a Blessing or a Curse ?

A Direct Measure of Transfer Effects [Longpre et al., 2025]



- English, French: very positive BTS scores across the board
- Strong positive effects of typological linguistic relatedness
- BTS scores are not fully symmetrical

Understanding the New Multilingual Abilities of PLMs

How can mPLMs handle Truly Multilingual Texts ?

- Input and **output** code-switched content?

Et le premier ministre nous répond que *a farmer is a farmer a Canadian is a Canadian* d'un bout à l'autre du Canada.

Autrement dit *they are getting out of the closet* parce que cela leur donne le droit d'avoir deux enfants.

(Exemples from [Carpuat, 2014])

- Generate texts with **multilingual prompts**

Přeložit do angličtiny: Ça n'arrête pas, nous sommes bien placés pour le savoir... ;

It doesn't stop, we know that well...

Answer questions X-linguistically

Generate summaries from **multilingual sources**

Understanding the New Multilingual Abilities of PLMs

From Language to Latent and Back: Observing Internal Alignments

Inspecting latent representations with early unembedding: the “Logit Lens”

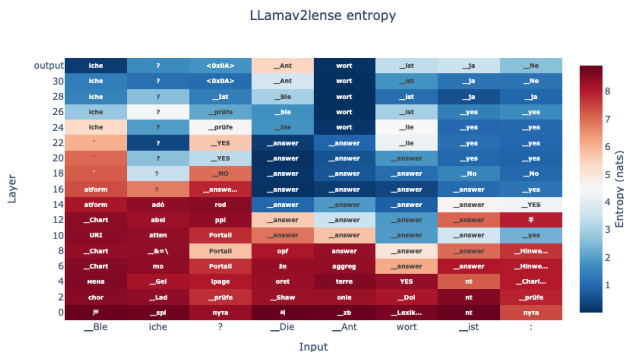


Figure from [Wendler et al., 2024]

Understanding the New Multilingual Abilities of PLMs

From Language to Latent and Back: Observing Internal Alignments



- latents reps align to English, **irrespective of the input language**
- the correct answer appears before the correct language
- in-language generation happens in the last few layers
- complementary evidences in [Schut et al., 2025]
- “**language specific neurons**” mostly in lower and upper layers [Tang et al., 2024]

Understanding the New Multilingual Abilities of PLMs

Computing parallel sentence alignment score across layers

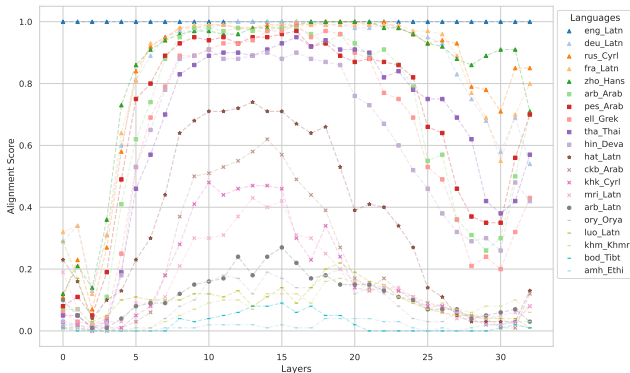


Figure from [Kargaran et al., 2025]

- alignment is predictive of transfer success
- improving alignment boosts X-lingual abilities

Part IV

Improving mPLMs

Monolingual Tokenization is almost Compression

Byte Pair Encoding (BPE): N deterministic merge operations

- 1 Make symbol map (**optimize compression**, greedy)
Repeat till done: merge most frequent bigram into new compound symbol (“tokens”)
- 2 Encode (greedy)
apply merge rules in training order - splits word in tokens

Monolingual Tokenization is almost Compression

Byte Pair Encoding (BPE): N deterministic merge operations

- 1 Make symbol map (**optimize compression**, greedy)
Repeat till done: merge most frequent bigram into new compound symbol (“tokens”)
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apply merge rules in training order - splits word in tokens

Example from [Sennrich et al., 2016]

$L = \{ \text{lower, lowest, newer, wider, wide} \}$

e	r#	3	[er#]		[lo]	w	2	[low]
l	o	2	[lo]		w	i	2	[wi]

Segmentations: [low]+[er#], [low]+e+s+[t#], n+e+w+[er#], [wid]+[er#],
[wid]+[e#]

🔗 <https://github.com/rsennrich/subword-nmt>

Monolingual Tokenization is almost Compression

Byte Pair Encoding (BPE): N deterministic merge operations

- 1 Make symbol map (**optimize compression**, greedy)
Repeat till done: merge most frequent bigram into new compound symbol (“tokens”)
 - 2 Encode (greedy)
apply merge rules in training order - splits word in tokens
- many implementations: YouTokenToMe, TkTkT, Tiktoken, SentencePiece, HuggingFace, etc.
 - BPE is training: **regularize** with sampling / pruning;
 - alternative models: SentencePiece (unigram model), wordPiece;
 - **pretokenization** (e.g., white space) matters
 - LM model $P(u_1, \dots, u_T | \theta)$, so that computing $P(w_1 \dots w_S | \theta)$ requires **summing over latent segmentations** [Cao and Rimell, 2021] or **avoid leaking probability** into **non-canonical segmentations** [Vieira et al., 2025].

Monolingual Tokenization is almost Compression

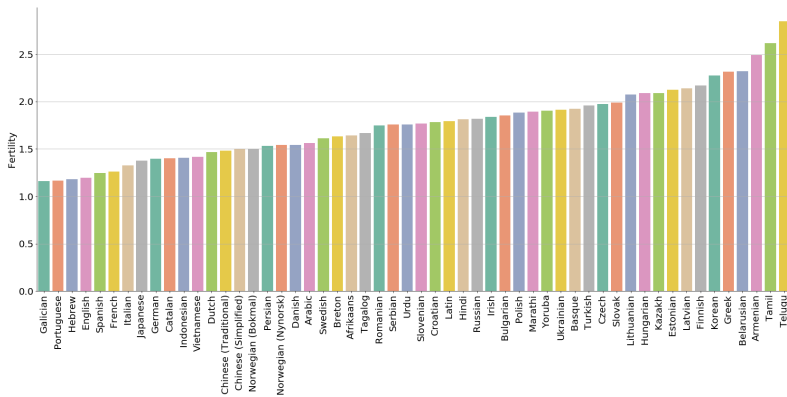
Byte Pair Encoding (BPE): N deterministic merge operations

- ① Make symbol map (**optimize compression**, greedy)
Repeat till done: merge most frequent bigram into new compound symbol (“tokens”)
- ② Encode (greedy)
apply merge rules in training order - splits word in tokens

Figures of Merits for a Good Tokenization

- small vocab is good: speeds up unembedding & softmax layer
- high compression is good: shortens input, speeds up encoding & decoding
 - **compression rate**: original length (in bytes) to the final length (in pieces)
 - **fertility**: average number of pieces / word (smaller is better)
- units that are frequently used:
 - high **average frequency**; low **Shannon entropy** of unigram distribution
 - high **type/token ratio**
 - high average **token rank**

Multilingual Tokenizers are Unfair

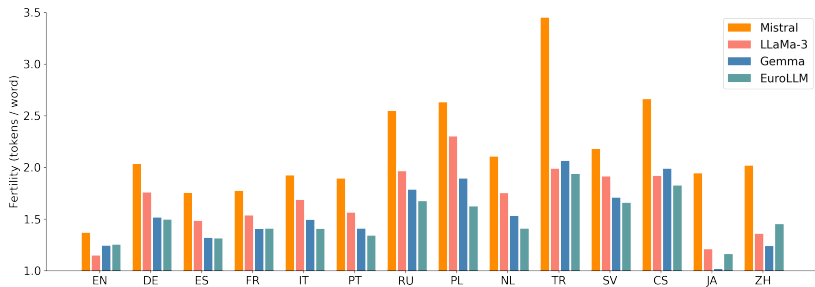


Fertility of mBERT tokenizer on UD datasets. HR languages have a better fertility, hence yield shorter sequences, and better processing costs [Ahia et al., 2023]

📊 Analysis by Judit Ács

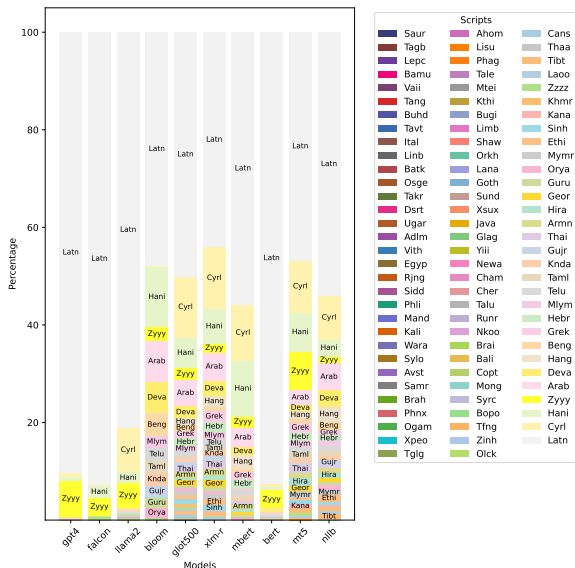
<https://juditacs.github.io/2019/02/19/bert-tokenization-stats.html>

Multilingual Tokenizers are Unfair



Fertility of various modern tokenizers for European Languages. Vocabulary sizes: 23k for Mistral; 128k for Llama-3 and EuroLLM; 256k for Gemma. [Martins et al., 2025]

Multilingual Tokenizers are Unfair



Most units are in **Latin script**
[Kargaran et al., 2024]

Multilingual Tokenizers are Unfair

BPE relies on simple statistics: bigram counts

- ⇒ **unbalanced training corpus** sizes across languages (higher diversity helps);
- ⇒ **unbalanced training distributions** for groups of related languages sharing script;
- ⇒ BPE is biased towards **morphologically simple**, synthetic languages (English, Chinese), also towards small letter inventories;
- ⇒ in the limit, **pretokenization helps morphologically complex** languages;

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- ⇒ in the limit, **pretokenization helps morphologically complex** languages;

Remedies:

- use more balanced language distributions (with squashing)
- allocate a fixed amount of units per language
- **parity-aware BPE** [Foroutan et al., 2025]
 - select next merge based on **statistics computed the least compressed language**
 - maximize compression rate of the worst language
 - requires **comparable corpora** for all languages

Boosting Alignment in Training

Standard Training Losses

$$\ell_{\text{MLM}}(\mathbf{x}) = \sum_{t \in \text{MASK}} -\log P(x_t | \mathbf{x}_{<t}; \boldsymbol{\theta}) \quad \text{Masked Language Model}$$

$$\ell_{\text{CLM}}(\mathbf{x}) = \sum_t -\log P(x_t | \mathbf{x}_{<t}; \boldsymbol{\theta}) \quad \text{Causal Language Model}$$

$$\ell_{\text{TM}}(\mathbf{x}, \mathbf{y}) = \sum_t -\log P(x_t | \mathbf{x}_{<t}, \mathbf{y}; \boldsymbol{\theta}) \quad \text{Causal Translation Model, } (\mathbf{y} \equiv \mathbf{x})$$

with:

$$-\log P(x_t | \mathbf{x}_{<t}; \boldsymbol{\theta}) = \text{sim}(x_t, \mathbf{x}_{<t}) - \log \sum_x \exp \text{sim}(x, \mathbf{x}_{<t}) \quad \text{Softmax layer}$$

where $\text{sim}(x_t, \mathbf{x}_{<t})$ is the dot product between the representation of x_t in the unembedding matrix, and the representation of $\mathbf{x}_{<t}$ on the output layer.

Boosting Alignment in Training

Some Auxiliary Training Losses – assuming parallel data ($\mathbf{y} \equiv \mathbf{x}$)

$$\ell_{\text{TLM}}(\mathbf{y}, \mathbf{x}) = \sum_{t \in \text{MASK}} -\log P(u_t | [\mathbf{y}, \mathbf{x}]_{-t}; \boldsymbol{\theta})$$

Masked Bilanguage Model

$$\ell_{\text{Dict-MLM}}(\tilde{\mathbf{y}}) = \sum_{t \in \text{MASK}} -\log P(x_t | \tilde{\mathbf{x}}_{-t}; \boldsymbol{\theta})$$

Masked CS Language Model

$$\ell_{\text{NCE}}(\mathcal{B}) = \sum_{i=1}^m \left(\frac{\text{sim}(\mathbf{y}^i, \mathbf{x}^i)}{\tau} - \log \sum_{\mathbf{u} \in \mathcal{B}, \mathbf{u} \neq \mathbf{y}^i} \exp \frac{\text{sim}(\mathbf{y}^i, \mathbf{u})}{\tau} \right)$$

Contrastive, sentencewise

- **TLM** [Conneau and Lample, 2019] resort to a simple language mixing scheme $[\mathbf{y}, \mathbf{x}]$;
- **DICT-MLM** use a dictionary to replace random words by their translation in $\tilde{\mathbf{x}}$.
- **NCE**: [Wei et al., 2021, Chi et al., 2021, Zhang et al., 2023], use a symmetricized version ($\mathbf{y}$ to $\mathbf{x}$, $\mathbf{x}$ to $\mathbf{y}$); also use a token-level version based on bilingual dictionaries.

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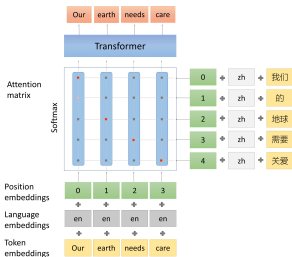
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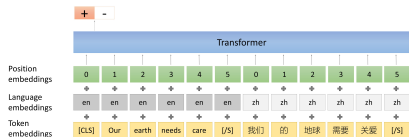
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Related: continuing training with **sentence-level objectives** to improve *sentence embeddings*: LABSE (from BERT) [Feng et al., 2022], GECKO [Lee et al., 2024], EMBEDDINGGEMMA [Vera et al., 2025]: sentence level classification tasks, retrieval, bitext mining

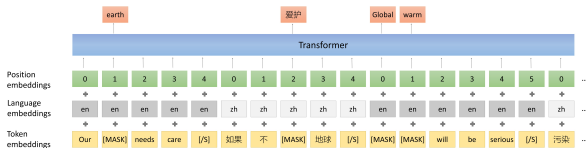
Boosting Alignment with Auxiliary Tasks



(a) Cross-lingual word recovery



(b) Cross-lingual paraphrase classification

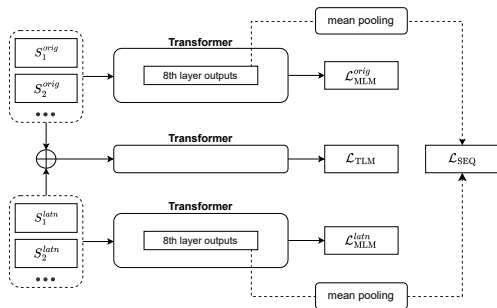


(c) Cross-lingual masked language model

Figure from [Huang et al., 2019]

Boosting Alignment with Data Preprocessing

- generate **artificial code-switching** data [Huang et al., 2019, Wei et al., 2021]
- include lots of **parallel data in late pretraining** [Martins et al., 2025];
- **transliterate** into a shared script [Muller et al., 2021];
- merge transliterated and original data + auxiliary losses [Xhelili et al., 2024].



$\ell_{MLM}^{orig, latn}$: MLM losse

$\ell_{MLM}^{orig, latn}$: TLM loss on original and romanized versions

ℓ_{SEQ} : sentence contrastive loss $\sim$ **NCE**)

Figure from [Xhelili et al., 2024]

Part V

Evaluating Multilingual PLMs

LLMs are Multitask Learners, their Evaluation is a serious matter

*In recent years, the AI technology that has arguably advanced the most is foundation models (Bommasani et al., 2021), (...) Yet the immense surface of model capabilities, limitations, and risks remains poorly understood. **The rapid development, rising impact, and inadequate understanding demand that we benchmark language models holistically.** [Liang et al., 2022]*

The multiple dimensions of LLM evaluation

As text generators	As representation learners
Good models of actual texts?	Semantic word / sentence similarities?
Discriminating well-formed utterances?	Easy to adapt / fine-tune ?
Generating realistic texts?	Useful for downstream applications?
Enabling controlled generation?	- in “ unstructured ” learners?
Generating factually correct texts?	- in “ structured ” learners?
Generating safe vs. toxic content?	Fair & unbiased wrt. user groups?

Also: Computationally and energy efficient? Generating accurate “reasoning”?

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Evaluating Text Generation with Distributional Metrics

An Alternative to Perplexity-based Evaluations

rep/ℓ: a repetition metric

Given a set $\mathcal{D}$ of length- T sequences,

$$\text{rep}/\ell = \frac{1}{|\mathcal{D}|T} \sum_{\mathbf{x} \in \mathcal{D}} \sum_{t=1}^T \mathbb{I}[\arg \max P(x_t | x_{<t}) \in \mathbf{x}_{t-\ell-1:t-1}].$$

Generalizes to repeated n-gram sequences.

Global distributional properties [Meister and Cotterell, 2021]

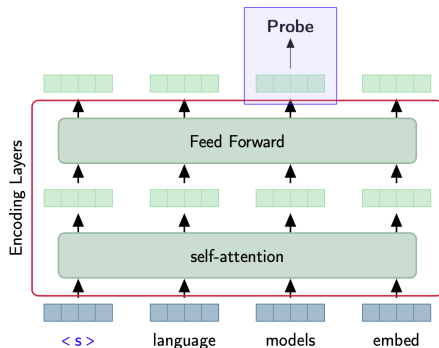
- Zipfian behavior, power-law distributions

$$P_{\text{zipf}}(W = w_k) \propto k^{-s}, s \approx 1$$

- **type-token ratios** (TTR) (depends on length)
- proportion of **hapax legomena** (frequency 1 words)
- proportion of specific lexical classes (punctuation, stopwords, etc)
- **consistency** metrics ?

Linguistic Probes

A common methodology to answer questions about the structure of models is to associate internal representations with external properties, by training a classifier on said representations that predicts a given property. This framework, known as **probing classifiers**, has emerged as a prominent analysis strategy in many studies of NLP models. [Belinkov, 2022]



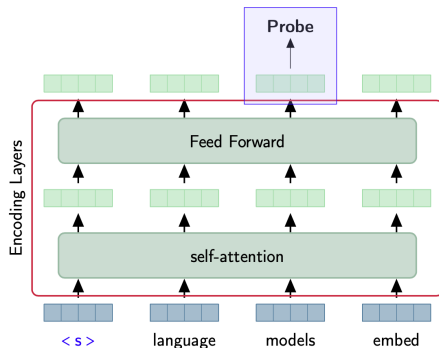
Probing Representations

- encoder outputs **contextualized representation** of input tokens
- linguistic probes: **supervised classifiers** to predict linguistic properties from the token representation
- small error rate $\Rightarrow$ **linguistic property encoded in representation**

Figure (C) G. Wisniewski

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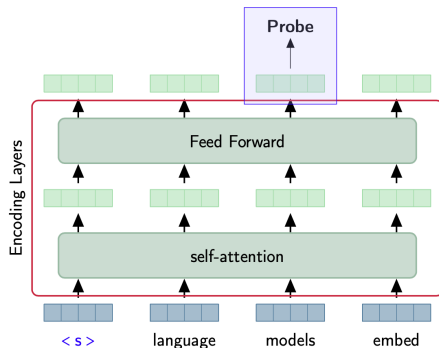
Word Level Probes

- what is the (gender, number, tense, mode, etc) of current token?
- what is the syntactic head of current token?
- what is the (gender, number, tense, mode, etc) of current token's head ?

Figure (C) G. Wisniewski

Linguistic Probes

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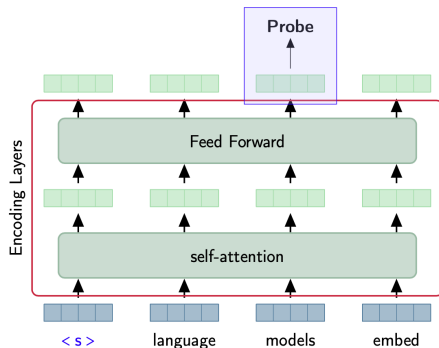
Sentences Level Probes via <s>

- what is the **sentence length**?
- does sentence **contain word X**?
- what is the (**gender, number, tense, mode, etc**) of sentence head?

Figure (C) G. Wisniewski

Linguistic Probes

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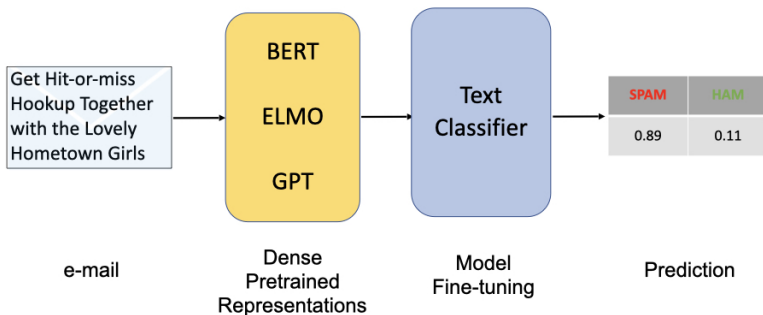
Caveats

- use **weak** classifiers and contrast with **random** labels
- how good should classification accuracy be ?
- a feature is learned does not mean a feature is used

Figure (C) G. Wisniewski

Using Representations in Downstream Tasks

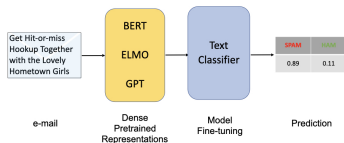
Extrinsic Evaluations in Classification Tasks



Improving spam filtering with pre-trained representations

Using Representations in Downstream Tasks

Extrinsic Evaluations in Classification Tasks



Task families, structured and unstructured

- **text mining**: text classification, sentiment analysis, hate speech detection, textual entailment, etc
- **linguistic tasks**: POS tagging, dependency parsing, named entity recognition (NER), etc

Figure of merits

- Performance on the task
- Trade-off **tuning effort vs. gains**

Using Representations in Downstream Tasks

Extrinsic Evaluations in Classification Tasks

Larger models, larger benchmarks

- **SUPERGLUE (10 tasks)**: a monolingual dataset for representation learning (10 tasks) [Wang et al., 2019]
- Tons of QA datasets *“Question answering and reading comprehension have been particularly prolific in this regard, with more than 80 new datasets appearing in the past 2 years.”* [Rogers et al., 2023]
- **BLIMP (67 datasets)**: (...) *challenge set for evaluating the linguistic knowledge of language models (LMs) on major grammatical phenomena in English* [Warstadt et al., 2020]
- **BIGBENCH (204 tasks)**: (...) *drawing problems from linguistics, childhood development, math, common-sense reasoning, biology, physics, social bias, software development, and beyond.* [Srivastava et al., 2023]
- **MTEB (1000 tasks)**: (...) *MTEB covers more than a 1000 different tasks from covering a diverse set of tasks from historic Swedish patent classification to documentation retrieval for Python* [Muennighoff et al., 2023]

Evaluating zero-shot / few-shot behaviour

Reduce NLP tasks to text generation with appropriate NL instructions as prompts

Prompts = instructions in Natural Language + [tricks] (from [Brown et al., 2020])

Specifically, we evaluate GPT-3 on over two dozen NLP datasets,(...) For each task, we evaluate GPT-3 under 3 conditions:

- “**zero-shot**” learning, where no **demonstrations** are allowed and only an instruction in natural language is given to the model.

Eval 125 + 12 =

- “**one-shot learning**”, where we allow only **one demonstration**, and

Eval 17 + 301 = 318 </s> Eval 125 + 12 =

- “**few-shot learning**”, or “in-context learning”, where we allow **as many demonstrations as will fit** into the model’s context window,

Eval 17 + 301 = 318 </s> Eval 48 + 67 = 105 </s> Eval 125 + 12 =

Tricks: “On tasks with free-form completion, we use beam search with the same parameters as [RSR+19]: a beam width of 4 and a length penalty of $\alpha = 0.6$.” (+ stopping criterion)

Evaluating zero-shot / few-shot behaviour

Reduce NLP tasks to text generation with appropriate NL instructions as prompts

Task types and their evaluation

Assuming prompt / instruction: $w_1 \dots w_T$.

- Yes / No answers

Question: *[Question]* True or false? *[prediction]*

Correct if $P(\text{True} | \text{prompt}) > P(\text{False} | \text{prompt})$.

- Question-Answering with multiple choice answers

Question: *Which factor will most likely cause a person to develop a fever?*

Correct Answer	<i>a bacterial population in the bloodstream</i>
Incorrect Answer	<i>a leg muscle relaxing after exercise</i>
Incorrect Answer	<i>several viral particles on the skin</i>
Incorrect Answer	<i>carbohydrates being digested in the stomach</i>

Correct if $P(\text{Correct answer} | \text{prompt}) > P(\text{Alternative} | \text{prompt})$

- One word continuation. Correct if $w_{T+1} == w^*$
- Multiple word continuation. Measure $\Delta(w_{T+1} \dots w_{T+S}; w_1^* \dots w_L^*)$ with $\Delta()$ task-dependent distance (ROUGE for summarization, BLEU for MT, etc)

Evaluating zero-shot / few-shot behaviour

Reduce NLP tasks to text generation with appropriate NL instructions as prompts

Understanding “instruction learning” results

Should pay attention to:

- how much effort went into prompting?
- free generation or text infilling or multi-choice answers?
- how were alternatives selected / generated?
- how was search performed (greedy or beam)?
- how does generation stops?
- how many shots is few shots?

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Open / new issues

- generating / optimizing discrete / continuous prompts
- optimizing retrieval and demonstration examples
- training with prompts and meta-learning

mPLMs being Multilingual, Evaluation is **very serious**

The multiple dimensions of mPLM evaluation

As text generators	As representation learners
Good models of actual texts? [1]	Semantic word/sentence similarities? [3]
Discriminating correct sentences? [1]	Easy to fine-tune ?
Generating realistic texts? [2]	Useful for downstream applications? [4]
Enabling controlled generation? [2]	- in “unstructured” learners ?
Generating factually correct texts? [6]	- in “structured” learners ?
Safe wrt. toxic & harmful content? [1]	Fair & unbiased wrt. user groups? [1]

[0] Evaluate mLMs as monolingual language models

New questions:

- [1] equally accurate, safe, fair for all languages? proportional to the training data?
- [2] does this include accidental or voluntary code-switching?
- [3] how about cross-lingual word associations?
- [4] including cross-lingual transfer learning?
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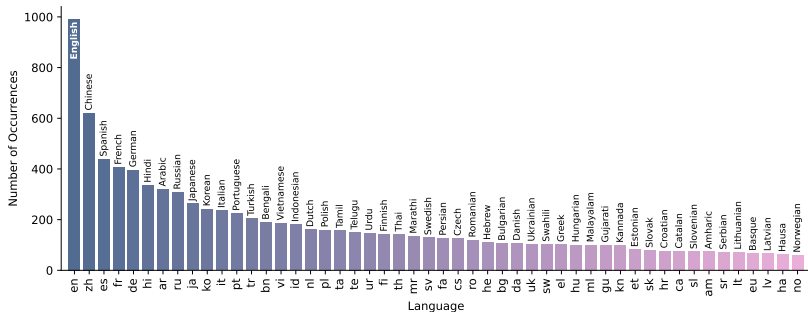
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LLMs as collections of Monolingual Models

Can they be evaluated at all? Language inequality strikes back, again !

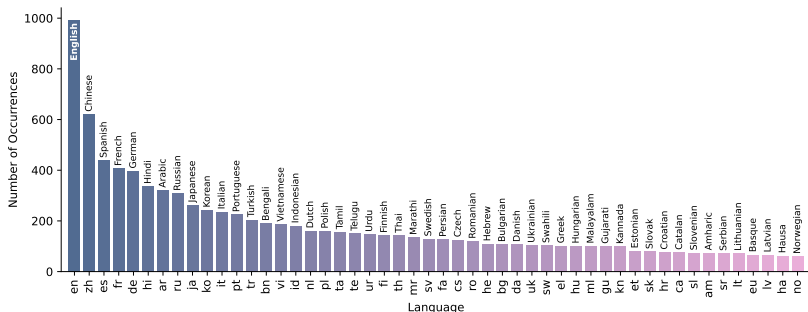


Benchmark development effort 2021-2024

Figure from [Wu et al., 2025]

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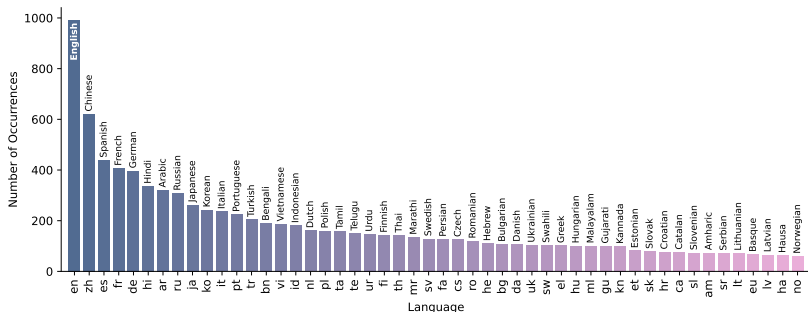
Figure from [Wu et al., 2025]

Can Machine Translated Benchmarks help?

- uneven performance across target languages / dialects / domains
- failing to account for cultural differences
- the impact of “translationese”

LLMs as collections of Monolingual Models

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Benchmark development effort 2021-2024

Figure from [Wu et al., 2025]

Can LLM-as-Judge replace annotations ?

- uneven performance across target languages / dialects / domains, detrimental to LR languages
- failing to account for cultural differences

A multi-lingual **meta-evaluation** in [Son et al., 2025]

LLMs as collections of Monolingual Models

Aggregation, Averaging and Comparison

Comparing / agregating scores across languages

- tasks may not compare in difficulty X-lingually
- scores may not compare in range / interpretation
BLEU count matches of **word n-grams**. Scores do not compare / average across target languages.

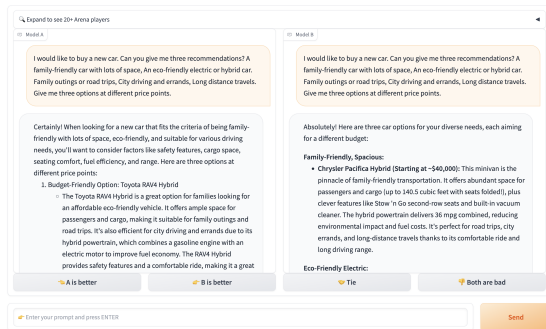
Alternative Aggregator- Borda Counts [Colombo et al., 2022]

- 1 rank models for each task / language from best to worst
- 2 aggregate with the sum / average **of ranks**

1,000,000 dollar question: how much are performance differences acceptable?

LLMs as collections of Monolingual Models

Can they be evaluated at all? Language inequality strikes back, again !



Our data involves more than 50 models, including both proprietary models like GPT-4, Claude, and Gemini, as well as open models such as LLaMA and Mistral. These conversations *cover more than 100 languages*, with 77% being in English, 5% in Chinese, and *the remaining languages, such as Russian, German, Spanish, French, and Japanese, each representing less than 2% of the total.*

Image and Quote from [Chiang et al., 2024]

The Perplexing Perplexity of Multilingual LMs

Questions that can be answered

Comparing **open world** models M_1 and M_2 perplexity-wise is fair if:

- ① performed with the exact same text $w_{1:T} = w_1 \dots w_T$
- ② computed with same normalizer (eg. T) irrespective of underlying segmentation / tokenization
- ③ $PPL(M_{\theta_i}) = P(w_{1:T} | \theta_i)^{-\frac{1}{T}} = 2^{-\frac{1}{T} \sum_{s=1}^S P(u_s | u_{<s}; \theta_i)}$

then:

$$PPL(M_1) < PPL(M_2) \Rightarrow M_1 \text{ better model than } M_2$$

The Perplexing Perplexity of Multilingual LMs

Questions that can be answered

For multilingual models,

- 1 $w_{1:T} = w_1 \dots w_T$ should be a multilingual text
- 2 and perplexity computed with the same normalizer irrespective of segmentation

Fair choice of $w_{1:T}$?

- Control for language / domain difficulty, e.g., using **multi-parallel** (Bible, Europarl) or **comparable** corpora to make all languages equally difficult (*).
- Control for language / domain unbalance

(*) caveat - this is notwithstanding translationese issues - translated texts are simpler? [Mielke et al., 2019]

The Perplexing Perplexity of Multilingual LMs

Questions that you **cannot** answer

Comparing perplexity scores across languages is risky

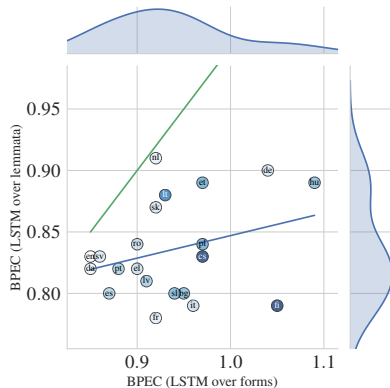
- 1 morphological divergences make lexical variation more or less severe:
5 forms / verb in English, ≈ 50 in French, thousands in Georgian
morphologically simple languages may be easier to predict
- 2 syntactic divergences make word order more or less fixed
fixed word order languages are easier to predict
- 3 using the **exact same text** is not possible - use translations instead?

Not all languages are equal

Some are intrinsically more “difficult-to-language-model” languages

The Perplexing Perplexity of Multilingual LMs

Questions that you **cannot** answer



Comparing language perplexity normalized per English characters for distinct LSTM-based language models. Figure from [Mielke et al., 2019]

The Perplexing Perplexity of Multilingual LMs

Questions that you **cannot** answer

Which language(s) are well or poorly language-modeled in a multilingual LM?

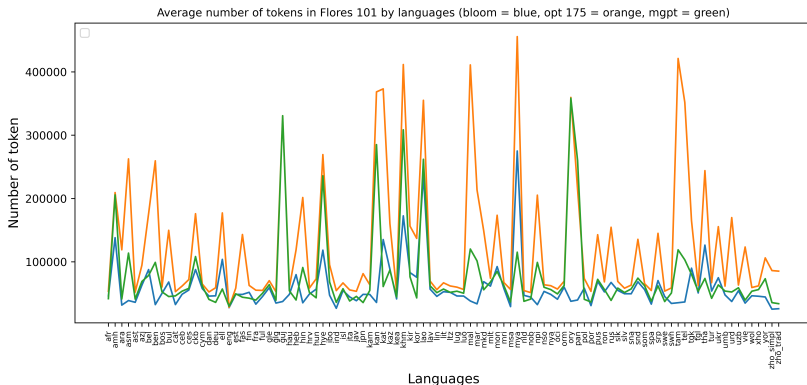
Experimental protocol

- 101 languages
- Flores multilingual test set, Wikipedia, translations from English [Goyal et al., 2022]
- 3 multilingual models:
 - BLOOM : 46 languages, byte level BPEs, 176B
 - mGPT: 60 languages, char level BPE, 13B
 - OPT: only English, byte level BPE, 175B

The Perplexing Perplexity of Multilingual LMs

Questions that you **cannot** answer

Which model is best language-modeled in a multilingual LM?

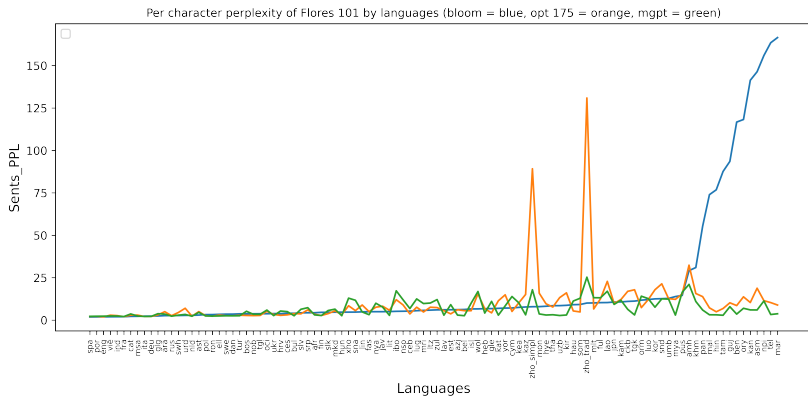


The variance of length: do not normalize per token.

The Perplexing Perplexity of Multilingual LMs

Questions that you **cannot** answer

How well is a language modeled in a multilingual LM?

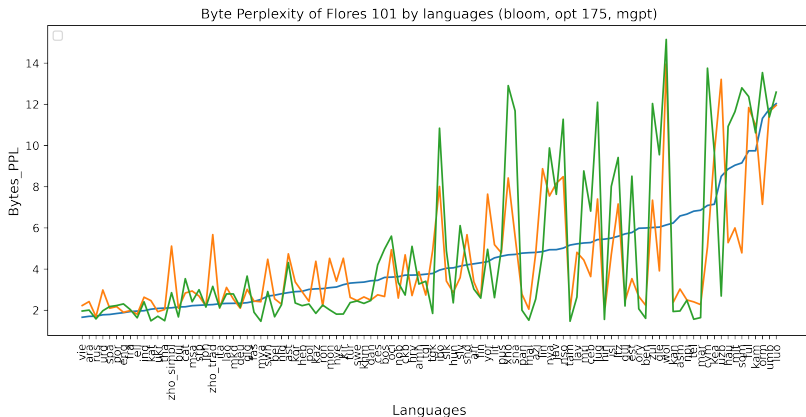


Normalizing per characters: improves PPL of languages with long words

The Perplexing Perplexity of Multilingual LMs

Questions that you **cannot** answer

Which model is best language-modeled in a multilingual LM??

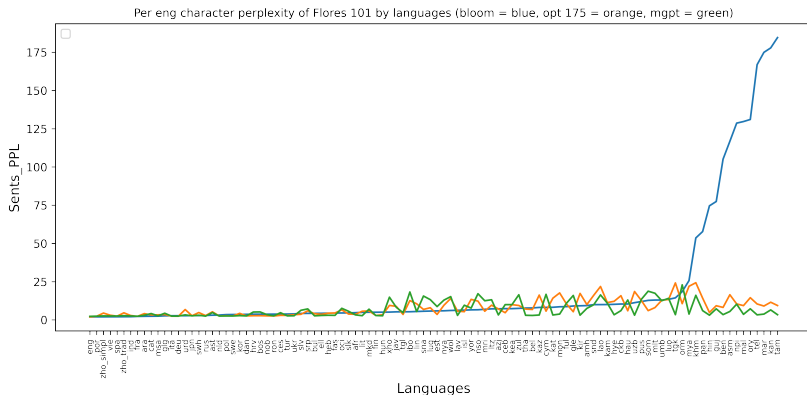


Normalizing per bytes: benefits languages using complex charsets

The Perplexing Perplexity of Multilingual LMs

Questions that you **cannot** answer

Which model is best language-modeled in a multilingual LM??



Normalizing per English characters: using less characters helps

Matching Parallel Sentences in Bitexts

Testing Multilingual Representation at Scale with Glot-500 [ImaniGooghari et al., 2023]

- unsupervised **word alignment** in parallel texts
- **parallel sentence mining** in multilingual corpora using x-sim (pct retrieval@1) scores

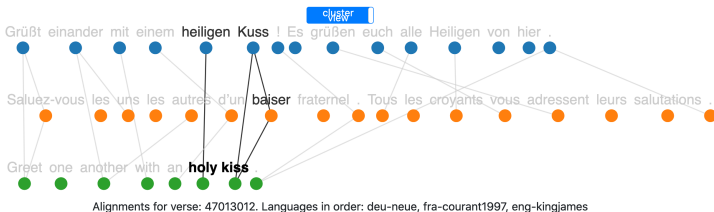
	lang-script	XLM-R-B	Glott500-m	gain		lang-script	XLM-R-B	Glott500-m	gain
Sentence Retrieval Tatoeba	tat_Cyrl	10.3	70.3	60.0	Sentence Retrieval Bible	uzn_Cyrl	5.4	87.0	81.6
	nds_Latn	28.8	77.1	48.3		crs_Latn	7.4	80.6	73.2
	tuk_Latn	16.3	63.5	47.3		srn_Latn	6.8	79.8	73.0
	ile_Latn	34.6	75.6	41.0		uzb_Cyrl	6.2	78.8	72.6
	uzb_Cyrl	25.2	64.5	39.3		bcl_Latn	10.2	79.8	69.6
	dtp_Latn	5.6	21.1	15.5		xav_Latn	2.2	5.0	2.8
	kab_Latn	3.7	16.4	12.7		mau_Latn	2.4	3.6	1.2
	pam_Latn	4.8	11.0	6.2		ahk_Latn	3.0	3.2	0.2
	lvs_Latn	73.4	76.9	3.5		aln_Latn	67.8	67.6	-0.2
	nob_Latn	93.5	95.7	2.2		nob_Latn	82.8	79.2	-3.6

English-X results from [ImaniGooghari et al., 2023]

Better Multilingual Representations improve Multilingual Sentence Retrieval

Matching Parallel Sentences in Bitexts

Testing Multilingual Representation at Scale with Glot-500 [ImaniGooghari et al., 2023]



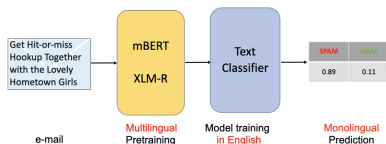
Evaluating Lexical Alignments with Round Trip Alignment

- 1 Compute unsup words alignment for all language pairs in **multiparallel corpus**
- 2 Repeat:
 - pick a sentence, K random languages
 - compute success rate of RT alignments $L_1 \rightarrow L_2 \dots L_1$

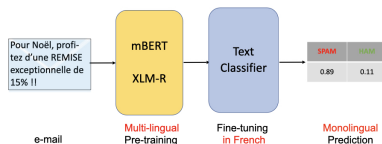
No gold annotation needed

Evaluating Representations through X-Lingual Learning

We adopt the zero-shot cross-lingual transfer setting, where we (1) fine-tune the pre-trained model on English and (2) directly transfer the model to target languages. [Conneau et al., 2020]



English spam filter



French spam filter

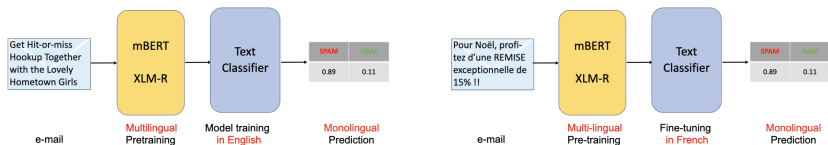
In “zero-shot” mode, no language adaptation is performed

Larger and wider benchmarks:

X-GLUE [Liang et al., 2020], X-TREME [Hu et al., 2020], BUFFET [Asai et al., 2024], X-TREME-UP [Ruder et al., 2023], Belebele [Bandarkar et al., 2024], INCLUDE [Romanou et al., 2024]

Evaluating Representations through X-Lingual Learning

We adopt the zero-shot cross-lingual transfer setting, where we (1) fine-tune the pre-trained model on English and (2) directly transfer the model to target languages. [Conneau et al., 2020]



English spam filter

French spam filter

In “zero-shot” mode, no language adaptation is performed

Larger and wider benchmarks:

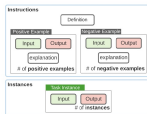
X-GLUE [Liang et al., 2020], X-TREME [Hu et al., 2020], BUFFET [Asai et al., 2024], X-TREME-UP [Ruder et al., 2023], Belebele [Bandarkar et al., 2024], INCLUDE [Romanou et al., 2024]

Evaluating Representations through X-Lingual Learning

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Evaluating Multilingual Instruction Following

76 types of tasks in 55 languages



SUPER-NATURALINSTRUCTIONS:

Generalization via Declarative Instructions on 1600+ NLP Tasks

Yizhong Wang², Swaroop Mishra³, Pegah Alipoormolabashi⁴, Yeganeh Kordi⁵, Amirreza Mirzaei⁶, Anjana Arunkumar⁷, Arjun Ashok⁶, Arut Selvan Dhanasekaran³, Atharva Naik⁷, David Stap⁷, Eshaan Pathak⁸, Giannis Karamanolakis¹⁰, Haizhi Gary Lai¹¹, Ishan Purubhi¹², Ishani Mondal¹³, Jacob Anderson³, Kirby Kuznia³, Krima Doshi³, Maitreya Patel¹³, Kuntal Kumar Pal³, Mehrad Moradshahi¹⁴, Mihir Parmar³, Mirali Purohit¹⁵, Neeraj Varshney⁵, Phani Rohitha Kaza³, Pulkit Verman¹⁶, Raveshaj Singh Puri¹, Rushang Karia³, Shulajha Keyur Sampat¹, Savan Doshi³, Siddhartha Mishra¹⁶, Suman Reddy¹⁷, Sumanta Patro¹⁸, Tanay Dixit¹⁹, Xudong Shen²⁰, Chitta Baral¹, Yejin Choi^{1,2}, Noah A. Smith^{1,2}, Hannaneh Hajishirzi^{1,2}, Daniel Khashabi²¹

¹Allen Institute for AI ²Univ. of Washington ³Arizona State Univ. ⁴Sharif Univ. of Tech. ⁵Bhara Polytechnic ⁶PG College of Tech. ⁷ITT Khargpur ⁸Univ. of Amsterdam ⁹UC Berkeley ¹⁰Columbia Univ. ¹¹Factored AI ¹²Govt. Polytechnic Rajkot ¹³Microsoft Research ¹⁴Stanford Univ. ¹⁵Zygon Informix ¹⁶Univ. of Massachusetts Amherst ¹⁷National Inst. of Tech. Karnataka ¹⁸TCS Research ¹⁹IFT Madras ²⁰National Univ. of Singapore ²¹Johns Hopkins Univ.

Task114: the given word

Definition: In this task, you need to answer 'Yes' if the given word is the longest word (in terms of number of letters) in the given sentence, else answer 'No'. Note that there could be multiple longest words in a sentence as they can have the same length that is the largest across all words in that sentence.

Input: Sentence: 'a man is surfing on a crashing wave.'. Is 'a' the longest word in the sentence?

Output: No

Input: Sentence: 'a man is riding on the back of an elephant'. Is 'is' the longest word in the sentence?

Output: No

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Slide (c) Philippe Langlais

Resources for Multilingual Evaluation

Some Leaderboards

- OpenEuroLLM Leaderboard
- European LLM Leaderboard
- Multilingual MMLU

Evaluation frameworks

- LM Harness by Eleuther AI
- Olmes
- Project MoonShot
- Deepeval / Confident AI
- ML Flows

Take Aways

Large mPLMs serve many practical purposes

They learn (universal ?) linguistic features

They display amazing emerging properties

Multilingual models also useful in MT, parsing, etc

They improve technological support for many languages [Ruder et al., 2023]

They require new tools - e.g., LID

They still remain poorly understood

- inference and transfer
- emergence of universal, language-agnostic features
- the effects of post-training

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